

Matrix Exponentials and Linear Systems

Math 5250 — Week 4

The Matrix Exponential

Definition. For a real or complex square matrix A , let

$$\exp(A) = e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

We can make sense of this using the Jordan normal form.

Reducing to Jordan Form

There is an invertible matrix P such that $B = PAP^{-1}$ is in Jordan form. Since $PA^kP^{-1} = B^k$,

$$P \exp(A) P^{-1} = I + B + \frac{B^2}{2!} + \frac{B^3}{3!} + \cdots = \exp(B).$$

So we can define $\exp(A)$ if we can make sense of $\exp(B)$.

The exponential respects the block structure of B , so it suffices to compute the powers of a single Jordan block.

Powers of a Jordan Block

Write a $k \times k$ Jordan block as

$$J_k(\lambda) = \begin{pmatrix} \lambda & & & & \\ 1 & \lambda & & & \\ & \ddots & \ddots & & \\ & & 1 & \lambda & \\ & & & 1 & \lambda \end{pmatrix} = \lambda I + N, \quad N^k = 0.$$

Since N commutes with λI ,

$$(\lambda I + N)^m = \sum_{j=0}^m \binom{m}{j} \lambda^{m-j} N^j.$$

The power N^j is obtained from N by shifting the ones “down and to the left”: it has ones on the j^{th} subdiagonal.

The Exponential of a Jordan Block

It follows that

$$e^{J_k(\lambda)} = e^\lambda \left(I + \sum_{j=1}^{k-1} \frac{N^j}{j!} \right),$$

and similarly

$$e^{tJ_k(\lambda)} = e^{\lambda t} \left(I + \sum_{j=1}^{k-1} \frac{t^j N^j}{j!} \right).$$

First-Order Linear Systems

Consider the system

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{pmatrix} = A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

where the x_i are functions of t and $\dot{x}_i = dx_i/dt$.

The solution is

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \exp(tA) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix},$$

where the a_i are the initial conditions (the values of the x_i at $t = 0$).

Solutions: Diagonal and Jordan Cases

If A is diagonal with entries $\lambda_1, \dots, \lambda_n$, the system breaks up into one-dimensional pieces with solutions

$$x_i = a_i e^{\lambda_i t}.$$

If $A = J_n(\lambda)$ is a Jordan block, then

$$x_1 = a_1 e^{\lambda t}$$

$$x_2 = e^{\lambda t}(a_2 + ta_1)$$

$$x_3 = e^{\lambda t} \left(a_3 + ta_2 + \frac{t^2}{2!} a_1 \right)$$

$$\vdots$$